

GitHub SSH Key Setup Guide

This guide explains how to configure an SSH key for GitHub so that you can clone, pull, and push repositories without entering your GitHub password each time.

1. Check for an Existing SSH Key

```
ls -al ~/.ssh
```

If you see files such as the following, you already have an SSH key:

```
id_ed25519
id_ed25519.pub
id_rsa
id_rsa.pub
```

GitHub recommends using an `ed25519` key for new setups.

2. Generate a New SSH Key

Replace the email address with the email used by your GitHub account:

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Press Enter to accept the default file location:

```
~/.ssh/id_ed25519
```

The command creates two files:

- `~/.ssh/id_ed25519`: private key. Do not share this file.
- `~/.ssh/id_ed25519.pub`: public key. This is the key you add to GitHub.

3. Start the SSH Agent

```
eval "$(ssh-agent -s)"
```

4. Add the Private Key to the SSH Agent

```
ssh-add ~/.ssh/id_ed25519
```

5. Copy the Public Key

```
cat ~/.ssh/id_ed25519.pub
```

Copy the full output. It usually starts with `ssh-ed25519`.

6. Add the SSH Key to GitHub

Open GitHub in your browser and go to:

```
Settings -> SSH and GPG keys -> New SSH key
```

Fill in the form:

- Title: a name that identifies this machine, such as `work Laptop`.
- Key: paste the public key copied from `~/.ssh/id_ed25519.pub`.

Click `Add SSH key`.

7. Test the GitHub SSH Connection

```
ssh -T git@github.com
```

If the setup is correct, GitHub will print a message like:

```
Hi username! You've successfully authenticated, but GitHub does not provide
shell access.
```

8. Use SSH URLs for Git Repositories

Clone a repository with an SSH URL:

```
git clone git@github.com:username/repository.git
```

For an existing repository that uses HTTPS, switch the remote URL to SSH:

```
git remote set-url origin git@github.com:username/repository.git
```

Check the current remote URL:

```
git remote -v
```